

Comparing the x86 to AMD Processor using SilverBench

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Abstract— This paper compares the x86 and AMD architectures, two of the most widely used processor architectures in modern computing. The comparison is based on various performance metrics such as power consumption, clock speed, instruction set, cache size, and multi-core capability. The comparison shows that while x86 processors are known for their high clock speed and single-core performance, AMD processors excel in multi-core performance and power efficiency. The paper also discusses the strengths and weaknesses of both architectures and their suitability for different applications.

I. INTRODUCTION

The x86 architecture, first introduced by Intel in the late 1970s, is one of the most widely used processor architectures in the world. It is found in desktop computers, laptops, servers, and even mobile devices. Over the years, the x86 architecture has undergone several revisions, with newer versions offering higher clock speeds, larger cache sizes, and improved instruction sets. In recent years, AMD has emerged as a strong competitor to Intel in the x86 market, with its Ryzen and Threadripper processors offering impressive multi-core performance and power efficiency.

The purpose of this paper is to compare the x86 and AMD architectures in terms of their performance, features, and suitability for different applications. The comparison is based on various benchmarks and performance tests, including power consumption, clock speed, instruction set, cache size, and multi-core capability. By comparing the strengths and weaknesses of both architectures, this paper aims to provide a comprehensive analysis of the two architectures and help readers make an informed decision about which architecture to choose for their computing needs.

II. LITERATURE REVIEW

Previous studies have compared x86 and AMD architectures, with some studies favoring one architecture over the other. For example, a study by Koc and Bayrakdar [1] found that AMD processors were more power-efficient than Intel processors, while a study by Zhang and Chen [2] found that Intel processors offered better single-core performance than AMD processors. However, these studies were conducted using older generation processors, and may not be representative of the latest processors from both companies.

III. METHODOLOGY

To compare the x86 and AMD architectures, we conducted a series of benchmarks using the SilverBench benchmarking tool. The benchmarks measured various aspects of processor performance, including power consumption, clock speed, instruction set, cache size, and multi-core capability. We used both single-core and multi-core benchmarks to compare the performance of both architectures in different scenarios.

IV. RESULTS

The results of the benchmarks showed that x86 processors had higher clock speeds and better single-core performance than AMD processors. However, AMD processors had better multi-core performance and were more power-efficient than x86 processors. The cache size of both architectures was similar, with some AMD processors having larger caches than some x86 processors. The instruction sets of both architectures were also similar, with both architectures supporting the same basic instructions.

V. CONCLUSION

In conclusion, this paper compared the x86 and AMD architectures in terms of their performance, features, and suitability for different applications. The comparison showed that while x86 processors had better single-core performance and clock speed, AMD processors had better multi-core performance and were more power-efficient. The paper also discussed the strengths and weaknesses of both architectures, and their suitability for different applications. Ultimately, the choice between x86 and AMD architectures will depend on the specific needs of the user,

such as whether they require single-core or multi-core performance, and whether power efficiency is a priority.

VI. ACKNOWLEDGEMENT

Thank you to Dr. Aly for teaching us about architectures. Thank you for the internet to help us find the sources we need to write this paper.

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